Riemann's result was that the properties of a curved space (or spacetime) in any dimension are described by a particular mathematical object, which we now call Riemann curvature and indicate with the letter 'R'. If you think of a landscape of plains, hills and mountains, the curvature *R* of the surface is zero in the plains, which are flat – 'without curvature' – and different from zero where there are valleys and hills; it is at its maximum where there are pointed peaks of mountains, that is to say, where the ground is least flat, or most curved. Using Riemann's theory, it is possible to describe the shape of curved spaces in three or four dimensions.

With a great deal of effort, seeking help from friends better versed in mathematics than himself, Einstein learns Riemann's maths – and writes an equation where R is proportional to the energy of matter. In words: spacetime curves more where there is matter. That is it. The equation is the analogue of the Maxwell equations, but for gravity rather than electricity. The equation fits into half a line, and there is nothing more. A vision – that space curves – becomes an equation.

But within this equation there is a teeming universe. And here the magical richness of the theory opens up into a phantasmagorical succession of predictions that resemble the delirious ravings of a madman but which have all turned out to be true. Even up to the beginning of the 1980s, almost nobody took the majority of these fantastical predictions entirely seriously. And yet, one after another, they have all been verified by experience. Let's consider a few of them.

To begin with, Einstein recalculates the effect of a mass like the Sun on the curvature of the space that surrounds it, and the effect of this curvature on the movements of the planets. He finds the movements of the planets as predicted by Kepler's and Newton's equations, but not exactly: in the vicinity of the Sun, the effect of the curvature of space is stronger than the effect of Newton's force. Einstein computes the movement of Mercury, the planet closest to the Sun and hence the one for which the discrepancy between the predictions of his and Newton's theories is greatest. He finds a difference: the point of the orbit of Mercury closest to the Sun moves every year 0.43 seconds of arc more than that predicted by Newton's theory. It is a small difference, but, within the scope of what astronomers were able to measure, and comparing the predictions with the observations of astronomers, the verdict is unequivocal: Mercury follows the trajectory predicted by Einstein, not the one predicted by Newton. Mercury, the fleet-footed messenger of the gods, the god of the winged sandals, follows Einstein, not Newton.

Einstein's equation, then, describes how space curves very close to a star. Due to this curvature, light deviates. Einstein predicts that the Sun causes light to curve around it. In 1919 the measurement is achieved; a deviation of